MATERIALS SCIENCE 2

MAGNETIC AND OPTICAL PROPERTIES OF MATERIALS

INTRODUCTION TO WEEK 13

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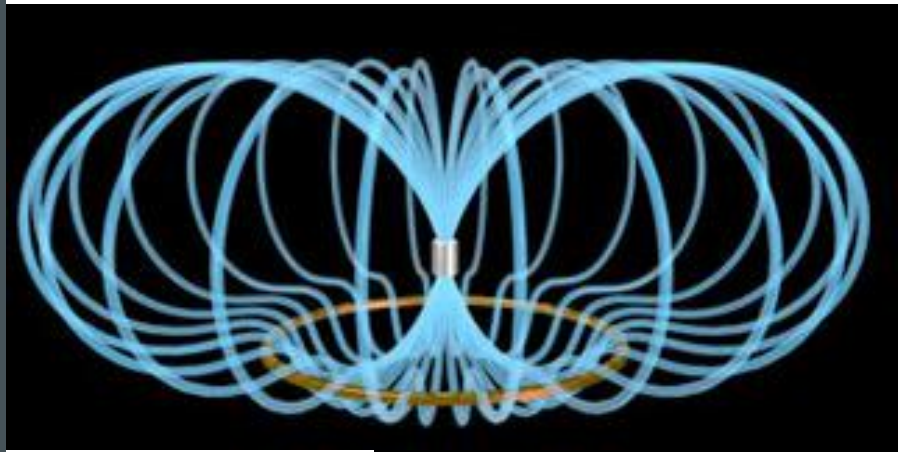

AIMS FOR THIS WEEK

Introduce	Discuss	Describe	Outline
Introduce the student to the mechanism that explains the magnetic behaviour of materials	Discuss the concepts of paramagnetism and ferromagnetism	Describe the optical properties of materials depending on their classification	Outline the mechanism of optical behavior of material

LEARNING OUTCOMES

- Explain the sources of magnetic moments in materials from an atomic perspective
- Explain the interaction of materials with light from an atomic perspective
- Determine the magnetization of a material using relevant equations
- Compare materials using their optical properties

WEEK OUTLINE

Monday	Live stream of lecture
Wednesday	Live stream of lecture

SKILLS

- Explain magnetic and optical properties of materials using atomic approach
- Ability to describe complex concepts using diagrams
- Listening technical English
- Perform calculations and analyse results
- Develop awareness of learning progress

COMMUNICATION IN THIS MODULE

Wecom Work groups: during class. This is not monitored outside class time

QMPlus forum: any question outside class time

Email: personal issues, not related to homework, class material, module content